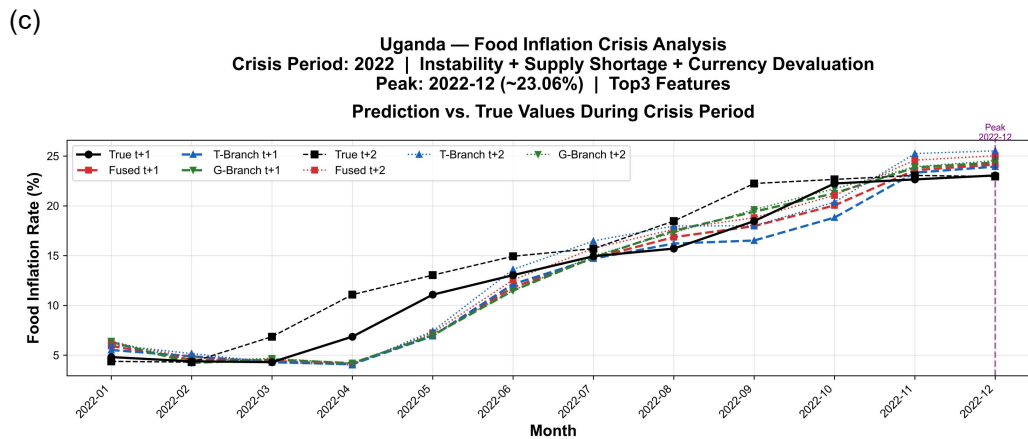
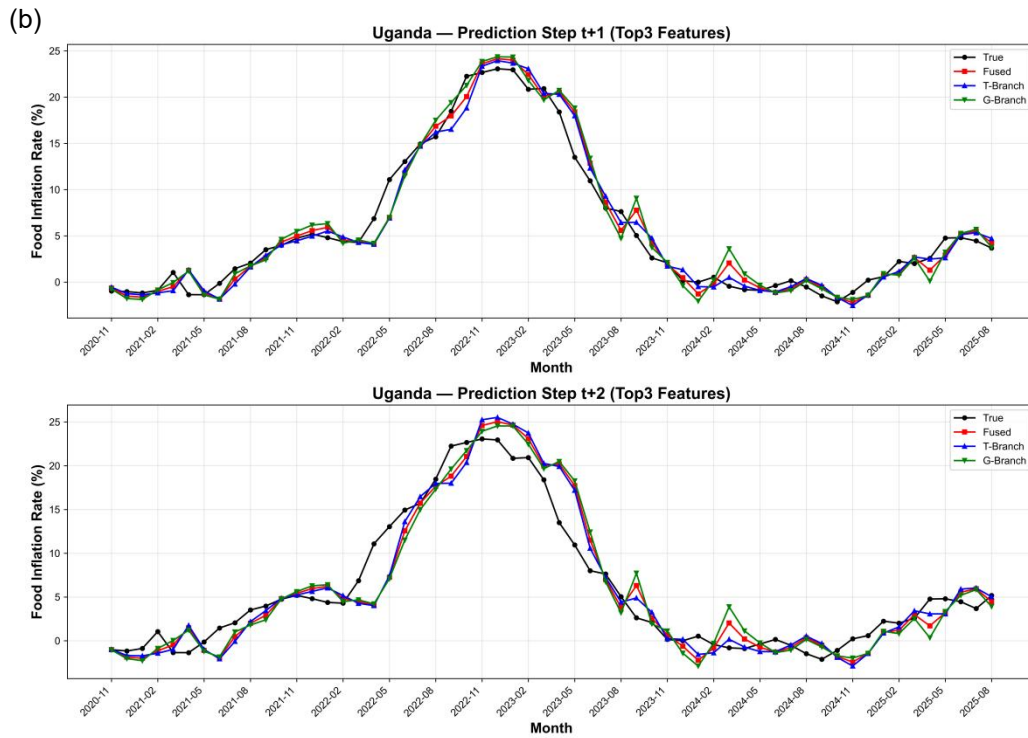
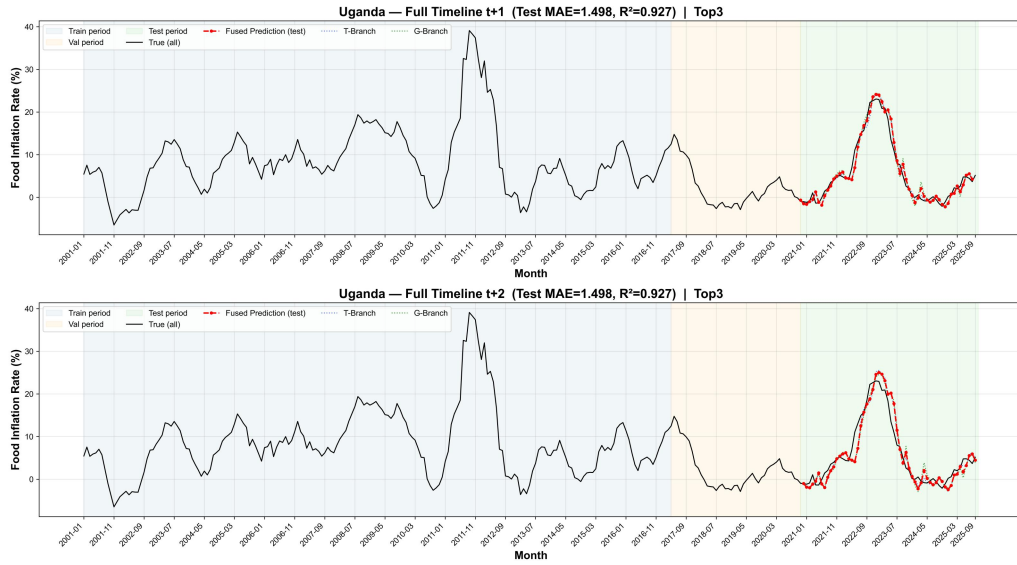
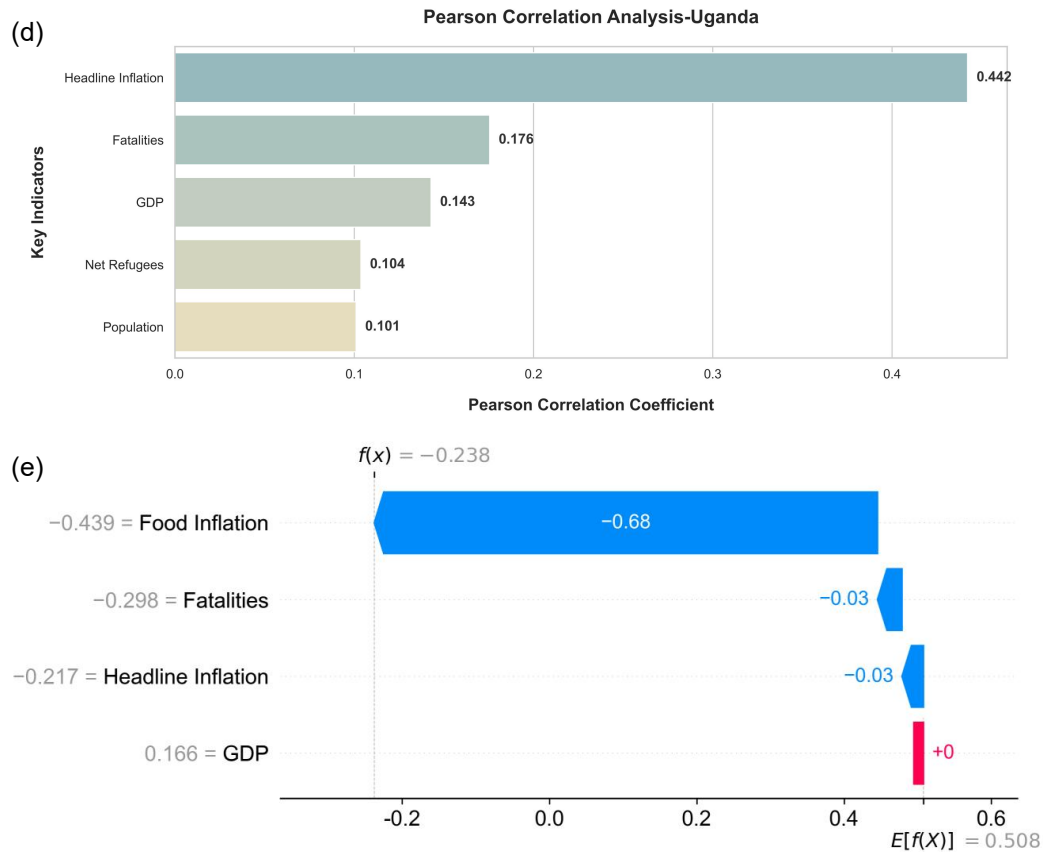






Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in Uganda. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

As a regional agricultural hub, food inflation in Uganda is primarily driven by robust export demand from neighboring countries, coupled with rising international input costs. Serving as a vital breadbasket for East Africa, Uganda sees its domestic retail benchmarks surge when supply deficits in neighboring Kenya or South Sudan drive high cross-border profit margins, triggering large-scale outflows of staples. On the production side, Uganda's smallholder farmers are acutely vulnerable to fluctuations in global fertilizer and fuel prices, and rising input costs frequently force agricultural contraction or elevated ex-farm prices. Despite a strong foundation for self-sufficiency, the lack of robust storage and price stabilization mechanisms allows international price volatility to be amplified through regional demand channels. This inflation profile is characterized by export-induced demand pressure, further reinforced by escalating long-distance logistics costs under the influence of global energy prices.